

LEMIEUX, QC, Canada

WMO: 716160

Lat: 46.300N

Lon: 72.060W

Elev: 319

StdP: 14.53

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.2	-14.7	-28.1	1.2	-19.7	-21.8	1.7	-13.3	21.8	23.5	19.4	23.2	2.3	180	0.581

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.2	84.8	70.7	82.0	68.8	79.3	67.3	73.7	81.1	71.8	78.2	70.0	76.4	9.7	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.3	116.9	77.7	69.6	110.2	75.7	67.7	103.1	73.8	37.6	81.3	35.8	78.3	34.3	76.6	81.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.7	16.5	14.5	DB	-31.6	89.4	7.8	2.3	-37.2	91.0	-41.7	92.4	-46.1	93.7	-51.7	95.4	
			WB	-30.6	76.6	7.4	2.2	-35.9	78.3	-40.3	79.6	-44.5	80.8	-49.9	82.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	40.8	10.6	13.6	25.0	39.5	53.3	61.8	66.4	63.9	56.7	45.2	32.7	19.7	(6)
	DBStd	21.56	14.55	12.83	11.77	8.09	7.57	6.99	5.55	6.32	7.68	7.93	9.69	12.23	(7)
	HDD50	5074	1221	1019	776	327	49	3	0	2	22	193	523	939	(8)
	HDD65	9049	1686	1439	1240	765	372	144	49	96	269	615	970	1404	(9)
	CDD50	1731	0	0	0	12	150	356	510	434	222	44	3	0	(10)
	CDD65	230	0	0	0	0	8	47	93	63	19	0	0	0	(11)
	CDH74	2256	0	0	0	11	175	521	847	535	163	4	0	0	(12)
	CDH80	469	0	0	0	2	39	127	188	87	26	0	0	0	(13)
	WSAvg	6.5	6.9	7.1	7.5	7.4	6.8	6.0	5.7	5.3	5.6	6.4	6.7	6.6	(14)
	PrecAvg	43.4	3.2	2.6	2.9	3.1	3.7	4.0	4.6	4.3	3.7	4.1	3.4	3.8	(15)
(16) Precipitation	PrecMax	50.7	6.2	4.7	5.5	6.5	7.1	7.6	7.3	6.4	7.1	7.3	5.4	8.6	(16)
	PrecMin	35.5	1.5	1.2	1.0	0.9	1.8	1.7	1.9	2.4	2.2	1.4	1.0	1.4	(17)
	PrecStd	4.3	1.1	0.9	1.1	1.4	1.3	1.5	1.5	0.9	1.1	1.5	0.9	1.5	(18)
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.9	45.4	56.0	73.1	84.9	87.7	88.9	85.8	83.8	73.6	63.4	49.1	(19)
		MCWB	46.4	40.3	46.1	57.7	65.3	71.7	74.0	72.7	71.6	62.6	57.3	46.8	(20)
	2%	DB	41.1	40.0	48.9	65.2	78.8	83.8	84.7	82.4	78.6	67.8	56.1	42.3	(21)
		MCWB	39.1	36.2	41.4	51.5	61.0	69.1	71.8	69.8	68.5	58.4	51.5	40.9	(22)
	5%	DB	36.1	36.3	44.4	59.1	74.1	80.2	81.6	79.7	74.7	62.8	51.1	38.3	(23)
		MCWB	34.3	33.9	38.7	48.1	59.6	66.6	69.6	68.4	65.5	56.4	47.2	36.5	(24)
	10%	DB	31.6	33.4	40.6	53.6	69.2	76.3	78.8	76.9	70.8	58.3	46.3	35.1	(25)
		MCWB	30.3	31.5	36.2	44.5	57.0	64.7	68.0	67.1	63.9	53.3	43.3	33.7	(26)
	0.4%	WB	46.4	42.4	49.7	60.4	70.0	74.8	76.8	75.1	73.5	65.0	58.1	47.2	(27)
		MCDB	47.1	44.1	53.8	71.7	79.3	84.1	85.2	83.0	81.5	70.9	61.3	49.3	(28)
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	39.0	36.7	43.1	53.9	65.6	71.8	74.2	72.5	70.4	60.6	52.7	41.3	(29)
		MCDB	40.3	39.0	47.0	61.8	74.0	79.5	81.4	78.7	76.2	66.3	55.8	42.7	(30)
	5%	WB	34.0	34.2	39.3	49.3	62.1	69.1	72.1	70.7	67.9	57.1	47.6	36.9	(31)
		MCDB	35.3	36.1	43.6	57.8	69.9	76.7	78.6	76.8	73.3	61.7	50.2	38.2	(32)
	10%	WB	29.9	31.6	36.6	45.6	58.8	66.8	70.2	69.0	65.2	53.9	43.5	33.9	(33)
		MCDB	31.2	33.4	40.1	52.3	66.6	73.7	76.4	75.0	70.3	57.3	46.3	34.9	(34)
(36) Mean Daily Temperature Range	5% DB	MDBR	20.2	22.2	19.9	19.2	23.3	22.6	21.2	21.8	21.9	17.7	14.9	15.8	(35)
		MCDBR	20.4	20.7	22.7	30.6	31.9	27.7	24.3	24.4	25.2	25.3	22.0	14.3	(36)
		MCWBR	19.3	18.4	16.0	18.4	16.4	14.4	12.9	13.6	15.2	16.8	17.2	14.1	(37)
	5% WB	MCDBR	18.8	19.0	20.2	27.0	24.8	23.0	20.5	20.4	22.2	22.1	19.6	14.9	(38)
		MCWBR	18.5	17.2	15.0	17.8	15.2	13.3	11.7	12.4	14.5	16.1	16.8	14.7	(39)
(40) Clear-Sky Solar Irradiance	taub		0.262	0.282	0.315	0.343	0.367	0.406	0.412	0.399	0.368	0.332	0.304	0.282	(40)
	taud		2.341	2.260	2.296	2.395	2.393	2.275	2.273	2.314	2.394	2.497	2.490	2.343	(41)
	Ebn at Noon		268	283	289	290	285	273	270	269	268	264	251	244	(42)
	Edh at Noon		23	31	35	35	36	41	41	38	32	25	21	21	(43)
(44) All-Sky Solar Radiation	RadAvg		450	734	1135	1413	1612	1769	1748	1529	1205	702	416	305	(44)
	RadStd		63	87	97	137	169	122	158	82	101	58	53	28	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+200	+72
(47)	Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page